

PUBLIC FACING TECHNICAL GUIDE

Installing a Dual Operating System

Windows + Ubuntu Desktop dual-boot setup for modern UEFI computers

This guide walks a technician or experienced user through preparing Windows, creating Ubuntu installation media, installing Ubuntu alongside Windows, and validating that both operating systems boot correctly.

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Document Control

Ownership, classification, and revision history for this public release.

Field	Value	Notes
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v1.0	2026-05-01	GreyNOC Comms	Initial public draft.
v1.1	2026-05-02	GreyNOC Comms	Internal peer review; minor wording updates.
v1.2	2026-05-04	GreyNOC Comms	AI-slop review pass: corrected grammar in pre-installation table ("expose existing data loss" → "risk data loss"), tightened vague qualifiers ("safe amount", "normal install", "careful user"), removed redundant step copy in Step 5, refreshed installer wording for the redesigned Ubuntu installer, and re-skinned to GN-TPL-PUB-001 layout (cover, document control, TOC, header/footer chrome).

Public release reminder

This document contains only externally approved content. It excludes internal toolchains, customer environments, credentials, and operationally sensitive detail per the GreyNOC public release rules.

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01 – Purpose and Scope

Use this guide when a user wants Windows and Ubuntu installed on the same physical computer.

What this guide covers

- Installing Ubuntu Desktop alongside an existing Windows installation.
- Creating sufficient unallocated disk space.
- Booting from a USB installer on UEFI systems.
- Confirming that both Windows and Ubuntu start correctly.
- Basic troubleshooting for common boot and encryption problems.

Assumptions

- Windows 10 or Windows 11 is already installed.
- The computer uses UEFI firmware and a GPT partition layout.
- The user has administrator access in Windows.
- Important data is backed up before any partition changes.
- The installer used in this example is Ubuntu Desktop LTS.

GreyNOC safety rule

Never resize partitions or install a second operating system until important files are backed up and the Windows recovery key is available. Dual booting is routine, but partition and firmware changes can cause downtime if preparation is skipped.

Recommended user level

Comfortable following technical instructions and using Windows administrative tools.

Estimated effort

About 45–90 minutes for a typical installation, not including backup time.

02 – What Dual Boot Means

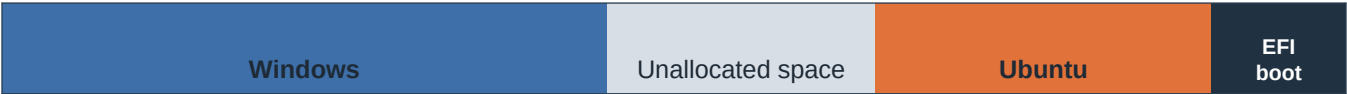
A dual-boot computer stores two operating systems on the same machine and lets the user choose one at startup.

How startup works

On modern computers, UEFI firmware starts the boot process. After Ubuntu is installed, the computer can present a boot menu where the user chooses Ubuntu or Windows Boot Manager. In many installations, Ubuntu's boot loader detects Windows automatically. If it does not, the user can still choose Windows from the firmware boot menu or repair the boot menu after installation.

Why install Windows first

When building a new dual-boot computer, install Windows first and Ubuntu second. Windows installers tend to assume they control the disk, while Ubuntu installers are designed to coexist with Windows. For an existing Windows laptop or desktop, leave Windows in place and create free space for Ubuntu.



Conceptual disk layout after a dual-boot installation. Sizes shown are illustrative.

Data separation

Keep work files backed up outside the laptop. A dual-boot setup can share files through a cloud service, an external drive, or a separate data partition, but each operating system needs enough disk space to update cleanly.

03 – Pre-Installation Checklist

Complete this checklist before booting into the Ubuntu installer.

Item	Action	Why it matters
Backup	Back up documents, photos, passwords, license keys, and recovery media.	Partition changes risk data loss or user error.
Power	Plug in laptops; avoid installation during storms or unstable power.	A shutdown mid-partition can corrupt the disk.
Recovery key	Save the BitLocker or Device Encryption recovery key outside the computer.	Firmware and boot changes can trigger a recovery prompt.
Space	Free at least 50 GB for Ubuntu; 80–120 GB is more comfortable.	Ubuntu needs space for the OS, apps, updates, logs, and user files.
USB drive	Use an 8 GB or larger USB drive. Its contents will be erased.	The Ubuntu ISO must be written as bootable media.
Internet	Have Ethernet or Wi-Fi credentials ready.	Updates and third-party drivers may need network access.

Do not skip the recovery key

If Windows Device Encryption or BitLocker is enabled, record the recovery key before changing boot settings, Secure Boot settings, or partitions. Store it in a password manager, in a printout, or in the Microsoft account recovery location.

Good USB label

Ubuntu Installer – May 2026

Good backup label

Pre-dual-boot backup – [date] – [device name]

Step 1 – Confirm Windows Boot Mode

Verify that the computer is using UEFI before changing partitions.

1 Open System Information

In Windows, press Windows + R, type msinfo32, and press Enter.

2 Check BIOS Mode

Look for **BIOS Mode**. For this guide it should read **UEFI**. Also note the Secure Boot state.

3 Check disk style when needed

Open Disk Management, right-click the main disk label, choose **Properties**, then **Volumes**. The partition style should normally be **GPT** on a modern UEFI Windows computer.

If the system uses Legacy BIOS or MBR

Stop and evaluate before proceeding. Mixing legacy and UEFI boot modes makes boot repair harder. A technician should plan a conversion, backup, or clean reinstall strategy before continuing.

Optional technician command

```
msinfo32
```

Step 2 – Prepare Windows Safely

Windows should be healthy and ready before Ubuntu is installed.

Recommended Windows prep

- Install pending Windows updates.
- Restart and confirm Windows starts cleanly.
- Free disk space using Storage settings if needed.
- Temporarily suspend BitLocker protection if it is enabled.
- Shut down fully before booting from USB.

Optional power setting

Disable Fast Startup so Windows volumes are fully closed when Linux accesses the disk.

```
powercfg /h off
```

Run Command Prompt as Administrator. Re-enable later with `powercfg /h on` if desired.

BitLocker handling

Suspending BitLocker does not decrypt the drive. It temporarily avoids recovery prompts while the boot environment changes. Resume protection after confirming Windows and Ubuntu both boot correctly.

Check BitLocker from an elevated terminal

```
manage-bde -status C:
```

Step 3 – Create Space for Ubuntu

Shrink the Windows partition from inside Windows and leave the new space unallocated.

- 1

Open Disk Management
Right-click **Start** and select **Disk Management**.
- 2

Shrink the Windows volume
Right-click the main Windows volume (usually **C:**), choose **Shrink Volume**, and enter the amount of space to free.
- 3

Leave it unallocated
Do not format the new space in Windows. The Ubuntu installer will create Linux partitions from the unallocated area.

Do not delete unknown partitions
EFI, Recovery, OEM, and Microsoft Reserved partitions may be required for startup or recovery. Only shrink the main Windows data volume unless a technician has a documented disk plan.

Ubuntu usage pattern	Suggested space
Testing, light browsing, basic terminal work	50–64 GB
Developer tools, Docker, labs, documentation	100–150 GB
Security labs, VMs, packet captures, large datasets	200 GB or more

Step 4 – Create the Ubuntu USB Installer

Download Ubuntu Desktop LTS and write it to a USB drive.

Download

Use Ubuntu Desktop LTS from the official Ubuntu download page. As of May 2026, Ubuntu 26.04 LTS is the current LTS release. The same workflow applies to future LTS versions.

Save the ISO file somewhere easy to find, such as Downloads.

Create USB media

Use a trusted bootable USB tool such as Rufus or balenaEtcher to write the ISO to the USB drive. Select the Ubuntu ISO, select the USB drive, and start the write process. The USB drive will be erased.

Recommended media notes

- Use a reliable USB drive of 8 GB or larger that has been tested.
- For UEFI systems, GPT/UEFI-compatible media is preferred.
- When Rufus asks about ISO mode or DD mode, ISO mode usually works for Ubuntu desktop installs.
- After writing, eject the USB drive cleanly.

Keep

Ubuntu ISO checksum when available.

Avoid

Unverified mirror links or modified installer images.

Step 5 – Boot the Ubuntu Installer

Start the computer from the USB drive instead of the internal disk.

- 1 Insert the installer USB**
Connect the installer USB to the target computer.
- 2 Open the boot menu**
Restart the computer and press the vendor boot-menu key. Common keys are **F12**, **F10**, **F9**, **Esc**, or **F2**.
- 3 Select the UEFI USB entry**
Choose the entry that references the USB device and UEFI. Avoid legacy entries when both are shown.
- 4 Try Ubuntu first if unsure**
The live environment lets you confirm keyboard, display, Wi-Fi, and trackpad support before installing.

Secure Boot note

Ubuntu supports Secure Boot on most common PCs. If the installer or installed system fails with Secure Boot errors, document the message before changing firmware settings. Some environments require Secure Boot to remain enabled.

Step 6 – Install Ubuntu Alongside Windows

Use the Ubuntu installer carefully and confirm the selected disk before writing changes.

- 1 Language and keyboard**
Choose the correct language and keyboard layout.
- 2 Network**
Connect to the internet if drivers and updates are needed.
- 3 Updates and software**
Normal installation is appropriate for most users. Minimal installation works for lighter systems.
- 4 Installation type**
Select **Install Ubuntu alongside Windows Boot Manager**, or the equivalent **Install alongside** option in the redesigned Ubuntu installer.
- 5 Manual option**
If using **Something else** or manual partitioning, work only inside the unallocated space created earlier.
- 6 User account**
Create a strong password and record the device name.
- 7 Review**
Read the final partition summary before clicking **Install**.

Partition warning

Do not select **Erase disk** unless the goal is to remove Windows. For a dual-boot installation, Windows partitions must remain intact.

Partition	Typical installer behavior
EFI System Partition	Existing EFI partition is reused for boot files.
Root filesystem	Ubuntu creates an ext4 (or similar) Linux filesystem in free space.
Swap	Modern Ubuntu typically uses a swap file instead of a separate swap partition.

Step 7 – First Boot Validation

Confirm that Ubuntu and Windows both start before returning the computer to normal use.

Ubuntu checks

- Ubuntu reaches the login screen.
- Keyboard, touchpad/mouse, display, and audio work.
- Wi-Fi or Ethernet connects.
- Software Updater or the App Center can reach repositories.
- The system clock is correct.

Windows checks

- Windows Boot Manager appears in the boot menu.
- Windows starts without recovery loops.
- BitLocker is resumed after validation.
- Important files are still present.
- Windows Update still works.

Common Ubuntu update commands

```
sudo apt update  
sudo apt upgrade
```

Document the final state

For client work, record the device model, Windows version, Ubuntu version, encryption state, boot menu behavior, and any firmware changes made during the install.

Troubleshooting Quick Reference

Use these checks for the most common post-install issues.

Symptom	Likely cause	First response
Computer boots straight to Windows	Firmware boot order favors Windows Boot Manager.	Open the UEFI boot menu and choose Ubuntu, then adjust boot order if appropriate.
BitLocker recovery prompt appears	Boot environment changed.	Enter the recovery key, boot Windows, verify health, then suspend or resume protection properly.
No Wi-Fi in Ubuntu	Driver or firmware not loaded.	Use Ethernet or USB tethering, run updates, then check Additional Drivers.
Wrong time in one OS	Windows and Linux handle the hardware clock differently.	Standardize time settings after both OSes boot reliably.
Secure Boot violation	Boot loader or firmware trust issue.	Record the exact error, check firmware settings, and validate installer media.
Installer cannot see Windows	Disk encryption, RAID/RST mode, damaged filesystem, or hibernated Windows.	Stop, boot Windows, back up, fully shut down, and re-evaluate storage controller settings.

When to stop

Stop troubleshooting and preserve evidence if the disk shows errors, Windows will not boot, recovery keys are missing, or the user reports irreplaceable data. Escalate to a technician with disk imaging and recovery experience.

GreyNOC Technician Handoff

Use this page to close out a client-facing dual-boot installation.

Device make/model		Windows edition/version	
Ubuntu version installed		Disk space assigned to Ubuntu	
BitLocker / Device Encryption status		Secure Boot status	
Firmware boot menu key		Known limitations or driver notes	

Client reminder

Your computer now contains two operating systems. Choose the one you need during startup. Keep backups current, apply updates in both systems, and do not delete partitions unless you are intentionally removing one of the operating systems.

Technician notes	Date

Appendix A – References and Source Notes

Primary references used to align this guide with official vendor guidance.

Ubuntu Desktop install tutorial Official Ubuntu guidance for booting from USB and installing Ubuntu Desktop. ubuntu.com/tutorials/install-ubuntu-desktop	Ubuntu release cycle Ubuntu release cadence and LTS versioning. Ubuntu 26.04 was released in April 2026. ubuntu.com/about/release-cycle
Microsoft Disk Management Microsoft description of Disk Management as the built-in Windows tool for creating, formatting, extending, and shrinking volumes. support.microsoft.com/windows/disk-management-in-windows	Microsoft BitLocker suspension guidance Microsoft guidance for suspending and resuming BitLocker protection around non-Microsoft updates or boot environment changes. learn.microsoft.com/troubleshoot/windows-client/windows-security/suspend-bitlocker-protection-non-microsoft-updates

Public guide disclaimer

This document is general technical guidance. Hardware vendors, firmware versions, encryption policies, and organizational security requirements vary. Follow client policy and vendor instructions where they differ from this general workflow.

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